

99_quickref

Quick Reference — per-milestone cheat sheet

M	Name	Tag	Source of truth to attach	The one test that proves it	Top hallucination risk
M0	Foundation/toolchain	Wk1	DBT_GUIDE §1, Bible §16–17	dbt debug green	wrong IAM role / dataset location not US
M1	Sources, macros, seed	Wk1	DBT_GUIDE §2–3, §6	macros compile; dbt seed	inventing event_params keys / a 2nd channel macro
M2	Staging	Wk1	DBT_GUIDE §3	dbt build --select staging	hallucinated raw GA4 columns
M3	Sessionization + funnel (KEYSTONE)	Wk1	DBT_GUIDE §4, DATA_MODEL §5	monotonicity + session_key uniqueness (golden, first)	did_* vs reached_* ; traffic_source as session source; wrong session_key
M4	Marts	Wk1	DBT_GUIDE §5, DATA_MODEL §3–8	revenue_reconciles to the cent; channels = 10	hallucinated columns; no transaction_id dedup
M5	Semantic layer live	Wk1	semantic_layer.yaml (exists), METRIC_GOVERNANCE_GUIDE	validate_semantic.py → 0 dangling refs	inventing metrics; wrong registry filename
M6	semantic-mcp + warehouse-mcp	Wk2	MCP_ARCHITECTURE §6.1–6.2, §9	build_query→dry_run→run_query→reconcile round-trip; budget gate	the LLM hand-writing SQL
M6b	stats / experiment / report MCP	Wk2 ¹	MCP_ARCHITECTURE §6.3–6.5, §9	decompose_change golden test	computing stats in prose
M7	Minimal loop (LEAN = 1 brief)	Wk2	AGENT_ARCHITECTURE §4, §6, §13	every brief figure traces to a tool output	the LLM inventing/recomputing numbers
M8	Memory	v2	Bible §22	save_diagnosis → recall_prior	—
M9	Full 7-agent loop	v2	AGENT_ARCHITECTURE §5–§10	FSM routing	—
M10	Eval harness (LEAN = 6–10, local)	Wk2	Bible §20, scenarios.yaml (exists)	Helios beats the naive baseline	over-claiming causal accuracy; an LLM-graded scorer
M11	Autonomy & depth	v2	Bible §23.3	scheduled run	autonomy theater on a frozen dataset
M12	Productionization/frontier	v2	Bible §23.4–5	—	—

¹ M6b in v1 = build only the **one** server you chose (semantic *or* stats); the rest are v2.

The golden rules (tape these to your monitor)

1. **CLAUDE.md at the root is your hallucination firewall** — never code without it loaded.
2. **No file without its source-of-truth doc section + the real upstream files attached.**
3. **Plan mode before any multi-file milestone.**
4. **Test-first** — and make Claude *run* the test and show you the output (don't trust, verify).

5. **The machine catches hallucinations** — `compile` + `dry_run` + `test` before you accept anything.
6. `/clear` between milestones.
7. **Numbers come from tools, not the LLM** (M7+) — the model narrates; `stats-mcp` computes.
8. **Spend LLM budget only on the brief** — build the deterministic layers with Claude Code.
9. **If Claude invents a name, that's a defect** — reject it, attach the registry, regenerate.

The honesty discipline (it's also a hallucination guard)

When Claude writes the M7 brief or the M10 eval writeup, make it state plainly what is *measured* vs *assumed*: the decomposition shows *where* a metric moved (attribution), not *why* (causation); the eval proves controlled-attribution accuracy vs a baseline, not real-world causal accuracy. Forcing this honesty prevents the model from generating confident causal stories the data can't support — the exact failure the red-team flagged.